

Hanwen Zhang

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EDUCATION

Sichuan University

Chengdu, China

B.Eng. in Computer Science and Technology; Honors Class

Sep 2024 – Expected Jun 2028

- **GPA: 3.96/4.00 · Rank: 1/12.**
- Relevant coursework: Theory of Optimization (98), Probability and Statistics (97), Artificial Intelligence (96), Multimodal Learning (90).

RESEARCH INTERESTS

Generative 3D vision and spatial intelligence, with interests in geometry-aware multimodal reasoning and world models for embodied agents.

RESEARCH EXPERIENCE

City University of Hong Kong

Jul 2026 – Present

Research Intern, Department of Data Science (Supervisor: Xinke Li)

Hong Kong

- Investigated sparse-view 3D scene reconstruction with video diffusion, predicting unobserved views from a small set of input images.
- Reproduced the GenRecon and ReconViaGen pipelines and diagnosed their failure modes through controlled evaluation of multi-stage reconstruction outputs.
- Analyzed how errors propagate from coarse structural prediction to downstream surface generation, with a focus on geometric consistency under limited observations.

Bone-to-Shape: Single-Image Animal Reconstruction

Jan 2026 – Sep 2026

Project Lead, Provincial Undergraduate Innovation Training Program

Sichuan University

- Built an end-to-end pipeline for single-image animal reconstruction using pose and shape priors, video-diffusion-based novel-view synthesis, and 3D Gaussian Splatting (3DGS).
- Investigated how anatomical and body-shape priors constrain geometry across generated views; first inventor on the Animal 3D Reconstruction Using Body-Shape Priors patent application 202611403381.7 (filed Sep 2026).

Deep-Hole Guardian: Embodied Perception for Defect Inspection

Jan 2026 – Sep 2026

Team Member, University-Level Undergraduate Innovation Training Program

Sichuan University

- Processed depth-camera point clouds for deep blind-hole defect detection and integrated perception outputs with SCARA robot inspection and control.

PUBLICATIONS

Yijia Chen, Ze Zhang, **Hanwen Zhang**, et al. “Kernel self-representation learning based fuzzy neighborhood outlier detection.” *International Journal of Approximate Reasoning*, 198:109769, 2026. [DOI](#). Third author; contribution: Methodology, Formal Analysis.

HONORS & AWARDS

First-Class Single-Category Scholarship, Sichuan University (2025–2026).

Third-Class Comprehensive Scholarship, Sichuan University (2024–2025).

Provincial Third Prize, Ascend AI Track, Huawei ICT Competition (2024).

Third Prize, Sichuan Division, Chinese Mathematics Competition (2025).

SKILLS

Methods: video diffusion, 3D Gaussian Splatting, multi-view geometry, point-cloud processing

Programming / ML: Python, PyTorch, C/C++, scikit-learn